AgriAssist: Multimodal Time-Series-Based Crop Pest Outbreak Prediction

Synopsis on
Deep Learning Project
[ICT-4442]

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Brief Description of the Project

Agricultural productivity and food security face severe challenges from sudden pest infestations driven by fluctuating agro-climatic conditions. This project presents AgriAssist, an end-to-end deep learning framework designed to predict crop pest outbreak risks by fusing multimodal agrometeorological records, satellite vegetation indices, and crop growth stages into sequential temporal models.

The system processes daily time-series features including ambient temperature, relative humidity, cumulative precipitation, Normalized Difference Vegetation Index (NDVI) reflectance values, and categorical crop growth stages (early, mid, flowering, harvest) to classify binary pest outbreak risks over a sliding temporal window. Four distinct model families are benchmarked: Multi-Layer Perceptrons (MLP) for non-linear tabular feature interactions, Long Short-Term Memory (LSTM) networks for recurrent temporal microclimate dependencies, 1D Convolutional Neural Networks (1D-CNN) for automatic temporal feature extraction, and Gated Recurrent
Units (GRU) for parameter-efficient gating dynamics. Models are benchmarked on Accuracy, Precision, Recall, F1-scores, and ROC-AUC over a temporal train-test split to optimize early warning sensitivity.

Dataset(s) Identified

Name: Agro-Meteorological and Crop Phenology Repository (AgriAssist Multi-Year Dataset)

Source: Open-Access Agro-Climatic and Satellite Repository

Details: Contains longitudinal daily meteorological and vegetation observations spanning 3+ years. Attributes include continuous weather parameters (temperature, humidity, rainfall), Normalized Difference Vegetation Index (NDVI) proxies, one-hot encoded crop phenological growth stages, and a binary target label (pest_outbreak: 0 = No Outbreak, 1 = Outbreak).

Proposed Models for Comparison

MLP Baseline: Multi-Layer Perceptron trained on flattened windowed feature representations for non-linear baseline benchmarking.

CNN Variant: 1D Convolutional Neural Network (1D-CNN) for localized cross-channel temporal feature extraction.

RNN / Sequence Model: Long Short-Term Memory (LSTM) network with dropout regularization to capture long-range climate dependencies.

Gated Sequence Model: Gated Recurrent Unit (GRU) network with adaptive gating to evaluate parameter efficiency and sequential tracking against standard LSTM performance.

Model-to-Member Assignment

Nihar P Shetty: Data Preprocessing, Feature Scaling & Multi-Layer Perceptron (MLP) Pipeline

Pranav Prabhu: 14-Day Sliding-Window Sequencing, LSTM Architecture & ROC-AUC Evaluation

Vineet Pai: Exploratory Data Analysis, 1D-CNN Model Tuning & Metrics Synthesis

Sunku Venkata Subhash: Temporal Tensor Batching, GRU Model Implementation & Ablation Studies

Existing State-of-the-Art Methods Related to the Proposed Project

Gradient-Boosted / Static Tree Models: Traditional tabular models (e.g., XGBoost, Random Forest, Shallow MLPs) operating on static aggregate weather statistics.

1D Convolutional Networks: Automatic feature extraction from sliding multimodal environmental windows.

Spatial-Temporal LSTMs: Recurrent sequence architectures that capture sequential dependencies from historical daily micro-climatic patterns.

Objectives

1. To construct and evaluate a comparative benchmark across tabular (MLP), convolutional (1DCNN), recurrent (LSTM), and gated sequential (GRU) deep learning architectures for binary agricultural pest outbreak forecasting using multimodal time-series data.
2. To quantify the predictive impact of integrating satellite vegetation health proxies (NDVI) alongside daily meteorological and crop growth indicators on classification F1-score and ROC-AUC.

Importance / Motivation of the Proposed Project

Pest infestations cause substantial pre-harvest agricultural yield losses worldwide. Traditional mitigation relies heavily on reactive, manual visual inspections, often resulting in delayed chemical interventions. Deploying automated, deep learning-driven early warning systems empowers farmers to apply precision agrochemicals proactively during critical initial vulnerability windows, reducing operational costs, minimizing ecological toxicity, and protecting crop yields.

Adapting deep learning pipelines to handle multimodal continuous time-series datasets addresses real-world constraints such as severe class imbalance, transition detection delay, and window sizing—providing a more realistic evaluation for deployment on mobile and edge IoT agroadvisory devices.

Preliminary Reference List

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5. R. S. Latha, B. R. Kavitha, and M. S. Kumar, "Multimodal Deep Neural Network for Crop Yield and Pest Infestation Prediction Using Meteorological and Satellite Indices," in *IEEE International Conference on Smart Information and Communication Technologies (SmartICT)*, pp. 245-250, 2021.

Work Plan

Sl. No.	**Component / Work Element / Milestone**	**Expected Completion (Day/ Month)**
1	Agro-meteorological dataset ingestion, cleaning, and exploratory data analysis (EDA)	30/08
2	Implementation of dynamic windowing, MinMax feature scaling, and one-hot encoding pipelines	05/09
3	Model implementation and baseline training (MLP, 1D-CNN, LSTM, GRU)	20/09
4	Hyperparameter tuning, ROC-AUC/F1 analysis on outbreak classes, and interim report completion	05/10